



Mathematics

4 Unit Assessment 15/12/1998

Question 1.

- (a) If $A = 3 + 4i$ and $B = 5 - 12i$ write the following in the form $a + ib$. (2 marks)
- (i) $A + B$
- (ii) AB
- (b) If $z = 2(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$ evaluate z^6 . (2 marks)
- (c) Find the square root of $5 - 12i$. (3 marks)
- (d) Draw neat labelled sketches (not on graph paper) to indicate each of the subsets of the Argand diagram described below : (16 marks)
- (i) $\text{Im}(z) = 4$
- (ii) $|z - 4| \leq |z + 4i|$
- (iii) $|z - 2 - 3i| = |z - i|$
- (iv) $|z| \leq 1$
- (iii) $|z - 3 + 4i| = 5$
- (iv) $\arg(z + i) = \frac{\pi}{4}$

Question 2.

- (a) Given that $z = \frac{1 - 7i}{3 + 4i}$, find (5 marks)
- (i) $|z|$
- (ii) z in the form $a + ib$
- (iii) $\bar{z}$
- (b) Find algebraically and describe in geometrical terms, the locus (in the Argand plane) represented by (6 marks)
- (i) $2|z| = z + \bar{z} + 4$
- (ii) $z\bar{z} = z + \bar{z}$
- (c) Find the modulus and argument of (8 marks)
- (i) $1 - i$
- (ii) $-\sqrt{3} - i$
- (iii) $(1 - i)(-\sqrt{3} - i)$
- (iv) $\frac{1 - i}{-\sqrt{3} - i}$
- (d) Let z be the complex number so that $z = 1 + \sqrt{3}i$ (5 marks)
- (i) Express z in the mod / arg form
- (ii) Write z^5 in the form $a + ib$.

Question 3.

- (a) Show that if $z = \cos \theta + i \sin \theta$ then
 $z^n + z^{-n} = 2 \cos n\theta$ (4 marks)

- (b) Use De Moivre's Theorem to show that
 $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$.

Hence show that the solutions of the equation $8x^3 - 6x + \sqrt{3} = 0$.

are $x = \sin \frac{\pi}{9}$, $x = \sin \frac{2\pi}{9}$ and $x = \sin \frac{13\pi}{9}$ (6 marks)

Show that $\sin \frac{\pi}{9} + \sin \frac{2\pi}{9} - \sin \frac{4\pi}{9} = 0$

- (c) Show that
 $\frac{1}{2}(1 + i\sqrt{3})(\cos \theta + i \sin \theta) = \cos\left(\frac{\pi}{3} + \theta\right) + i \sin\left(\frac{\pi}{3} + \theta\right)$ (3 marks)

- (d) What is the maximum value of $|z|$ for $|z - 1 - i| \leq 2$ (3 marks)

Question 4.

- (a) (i) Find the fifth roots of unity.
 (ii) Show these roots on an Argand diagram
 (iii) Show that the complex roots occur in conjugate pairs. (6 marks)
 (iv) If ω is one of the roots of $z^5 = 1$ state the value of
 $\omega + \omega^2 + \omega^3 + \omega^4$

- (b) Show by means of an Argand diagram, that if z and w are complex numbers then
 $|z + w| \leq |z| + |w|$ (3 marks)

- (c) On separate diagrams draw a neat sketch of the locus specified by each of the following .
 Include a discription to clarify your sketch where necessary.

(i) $0 \leq \text{Arg}(z + 1) \leq \frac{\pi}{4}$

(ii) $\left| \frac{z+1}{z-1} \right| = 1$ (6 marks)

(iii) $\text{Arg}\left(\frac{z+1}{z-1}\right) = \frac{\pi}{4}$

- (d) Simplify

$$\frac{(\cos 5\theta - i \sin 5\theta)^2 (\cos 3\theta + i \sin 3\theta)^8}{(\cos 2\theta + i \sin 2\theta)^5}$$
 (3 marks)

4 Unit Assessment

Q1 a) 15/12/98.

$$A = 3+4i, B = 5-12i$$

$$A+B = (3+4i) + (5-12i) \\ = 8-8i$$

$$AB = (3+4i)(5-12i) \\ = 15-36i+20i-48i^2 \\ = 15+48-16i \\ = 63-16i$$

$$b) Z^6 = \left[2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) \right]^6 \\ = 2^6 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)^6 \\ = 2^6 \left(\cos \left(\frac{\pi}{6} \times 6 \right) + i \sin \left(\frac{\pi}{6} \times 6 \right) \right) \\ = 64 \left(\cos \pi + i \sin \pi \right) \\ = 64(-1 + i \times 0) \\ = 64 \times -1 \\ = -64$$

$$c) x+iy = \sqrt{5-12i} \\ (x+iy)^2 = 5-12i \\ x^2+2ixy+i^2y^2 = 5-12i \\ x^2-y^2+2ixy = 5-12i \\ x^2-y^2=5 \text{ and } 2xy=-12 \\ x^2-y^2=5 \text{ and } y = \frac{-6}{x}$$

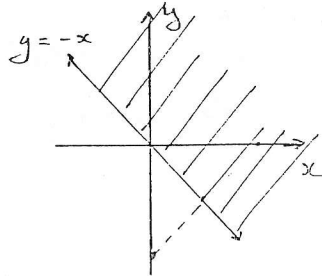
$$\therefore x^2 - \left(\frac{-6}{x} \right)^2 = 5 \\ x^2 - \frac{36}{x^2} = 5 \\ x^4 - 36 = 5x^2 \\ x^4 - 5x^2 - 36 = 0 \\ (x^2-9)(x^2+4) = 0 \\ x^2=9 \text{ OR } x^2=-4 \\ x = \pm 3 \text{ No solns as } x \text{ is real}$$

$$f) x=3, y = \frac{-6}{3} = -2 \\ x=-3, y = \frac{-6}{-3} = 2$$

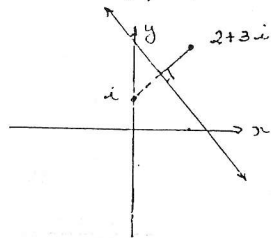
$$\therefore \sqrt{5-12i} = 3-2i \text{ OR } -3+2i \\ = \pm(3-2i)$$

$$d) i) \operatorname{Im}(z) = 4 \\ y = 4$$

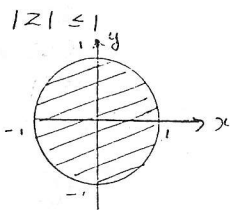
$$(ii) |z-4| \leq |z+4i|$$



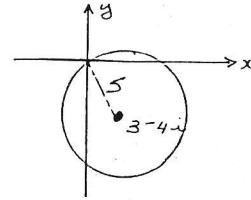
$$(iii) |z-2-3i| = |z-i| \\ |z-(2+3i)| = |z-i|$$



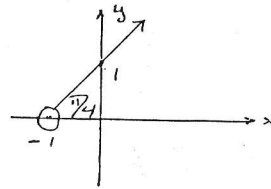
(iv)



$$v) |z-3+4i| = 5 \\ |z-(3-4i)| = 5$$



$$(vi) \arg(z+1) = \frac{\pi}{4}$$



Question 2

$$a) (i) z = \frac{1-7i}{3+4i} \times \frac{3-4i}{3-4i} \\ = \frac{3-4i-21i+28i^2}{3^2-16i^2} \\ = \frac{-25-25i}{25} \\ = -1-i$$

$$(ii) |\bar{z}| = \sqrt{(-1)^2 + (-1)^2} \\ = \sqrt{1+1} \\ = \sqrt{2}$$

$$(iii) \bar{z} = -1+i$$

$$b) (i) 2|z| = z + \bar{z} + 4$$

$$2\sqrt{x^2+y^2} = x+iy+x-iy+4$$

$$2\sqrt{x^2+y^2} = 2x+4$$

$$\sqrt{x^2+y^2} = x+2$$

$$x^2+y^2 = (x+2)^2$$

$$x^2+y^2 = x^2+4x+4$$

$$y^2 = 4x+4 \\ y^2 = 4(x+1)$$

Parabola vertex $(-1, 0)$
axis the x axis

$$(ii) z\bar{z} = z+\bar{z}$$

$$(x+iy)(x-iy) = x+iy+x-iy$$

$$x^2-y^2 = 2x$$

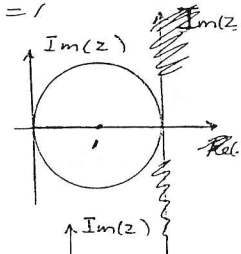
$$x^2+y^2 = 2x$$

$$x^2-2x+1+y^2 = 1$$

$$(x-1)^2+y^2 = 1$$

Circle
Centre $(1, 0)$

Radius 1



$$c) (i) 1-i$$

$$|1-i| = \sqrt{2}$$

$$\arg(1-i) = -\frac{\pi}{4}$$

$$(ii) -\sqrt{3}-i$$

$$|-\sqrt{3}-i| = 2$$

$$\arg(-\sqrt{3}-i) = -\frac{5\pi}{6}$$



$$(iii) |(1-i)(-\sqrt{3}-i)|$$

$$= |1-i| |-\sqrt{3}-i|$$

$$= \sqrt{2} \times 2$$

$$= 2\sqrt{2}$$

$$\arg\{(1-i)(-\sqrt{3}-i)\}$$

$$= \arg(1-i) + \arg(-\sqrt{3}-i)$$

$$= -\frac{\pi}{4} + (-\frac{5\pi}{6})$$

$$= -\frac{3\pi-10\pi}{12}$$

$$= -\frac{13\pi}{12}$$

$$= \frac{11\pi}{12}$$

$$(iv) \left| \frac{1-i}{-\sqrt{3}-i} \right| = \frac{|1-i|}{|-\sqrt{3}-i|}$$

$$= \frac{\sqrt{2}}{2}$$

$$\arg \left\{ \frac{1-i}{-\sqrt{3}-i} \right\} = \arg(1-i) - \arg(-\sqrt{3}-i)$$

$$= -\frac{\pi}{4} - -\frac{5\pi}{6}$$

$$= \frac{-3\pi + 10\pi}{12}$$

$$= \frac{7\pi}{12}$$

$$d) (i) z = 1 + \sqrt{3}i$$

$$= 2 \left\{ \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right\}$$

$$(ii) z^6$$

$$= 2^6 \left\{ \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right\}^6$$

$$= 2^6 \left\{ \cos \left(6 \times \frac{\pi}{3} \right) + i \sin \left(6 \times \frac{\pi}{3} \right) \right\}$$

$$= 2^6 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$$

$$= 2^6 \times \left(\cos \frac{2\pi}{3} - i \sin \frac{2\pi}{3} \right)$$

$$= 32 \left(\frac{1}{2} - i \frac{\sqrt{3}}{2} \right)$$

$$= 16(1 - i\sqrt{3})$$

Question 3:

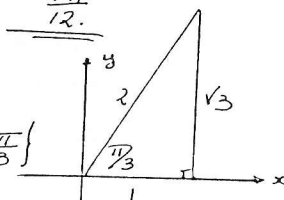
$$a) z^n + z^{-n}$$

$$= (\cos \theta + i \sin \theta)^n + (\cos \theta + i \sin \theta)^{-n}$$

$$= \cos n\theta + i \sin n\theta + \cos(-n\theta) + i \sin(-n\theta)$$

$$= \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta$$

$$= 2 \cos n\theta$$



$$b) (\cos \theta + i \sin \theta)^3 = \cos^3 \theta + 3 \cos^2 \theta i \sin \theta + 3 \cos \theta i^2 \sin^2 \theta + i^3 \sin^3 \theta$$

$$\cos 3\theta + i \sin 3\theta = \cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta$$

Equating imaginary parts

$$\sin 3\theta = 3 \cos^2 \theta \sin \theta - \sin^3 \theta$$

$$= 3(1 - \sin^2 \theta) \sin \theta - \sin^3 \theta$$

$$= 3 \sin \theta - 3 \sin^3 \theta - \sin^3 \theta$$

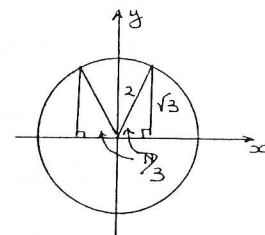
$$= 3 \sin \theta - 4 \sin^3 \theta$$

$$\text{let } \sin \theta = x \text{ and } \sin 3\theta = \frac{\sqrt{3}}{2}$$

$$\therefore \frac{\sqrt{3}}{2} = 3x - 4x^3$$

$$\sqrt{3} = 6x - 8x^3$$

$$8x^3 - 6x + \sqrt{3} = 0.$$



$$\therefore 3\theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{8\pi}{3}$$

$$\theta = \frac{\pi}{9}, \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}, \frac{8\pi}{9}$$

$$\therefore x = \sin \frac{\pi}{9}$$

$$x = \sin \frac{2\pi}{9}$$

$$x = \sin \frac{4\pi}{9}$$

$$= \sin \frac{5\pi}{9}$$

$$x = \sin \frac{7\pi}{9}$$

$$= \sin \frac{8\pi}{9}$$

$$x = \sin \frac{10\pi}{9}$$

$$= -\sin \frac{4\pi}{9}$$

$$x = \sin \frac{14\pi}{9}$$

$$= \sin \left(-\frac{4\pi}{9} \right)$$

$$= -\sin \frac{4\pi}{9}$$

$\therefore$ Roots are

$$x = \sin \frac{\pi}{9}, x = \sin \frac{2\pi}{9}$$

$$x = -\sin \frac{4\pi}{9}$$

$$\sum x = -\frac{b}{a}$$

$$= \frac{0}{8}$$

$$= 0$$

$$\therefore \sin \frac{\pi}{9} + \sin \frac{2\pi}{9} + (-\sin \frac{4\pi}{9}) = 0$$

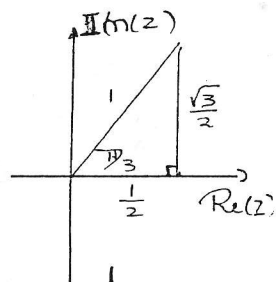
$$\sin \frac{\pi}{9} + \sin \frac{2\pi}{9} - \sin \frac{4\pi}{9} = 0.$$

$$c) \frac{1}{2}(1+i\sqrt{3})(\cos 0 + i \sin 0)$$

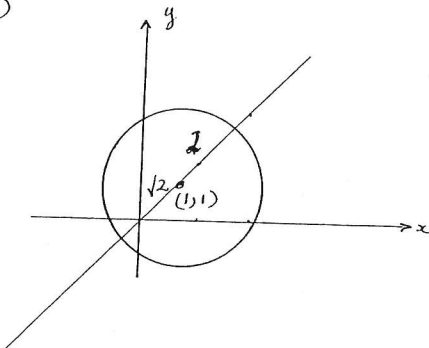
$$= 1\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right) \times 1(\cos 0 + i \sin 0)$$

$$= 1 \times 1 \left\{ \cos\left(\frac{\pi}{3} + 0\right) + i \sin\left(\frac{\pi}{3} + 0\right) \right\}$$

$$= 1 \left\{ \cos\left(\frac{\pi}{3} + 0\right) + i \sin\left(\frac{\pi}{3} + 0\right) \right\}$$



d)



Max. Value is $\sqrt{2} + 2$.

Question 4.

$$(i) z^5 = 1$$

$$\text{let } z = r(\cos \theta + i \sin \theta)$$

$$\{r(\cos \theta + i \sin \theta)\}^5 = 1(\cos 0 + i \sin 0)$$

$$r^5 (\cos \theta + i \sin \theta)^5 = 1(\cos(0 + 2n\pi) + i \sin(0 + 2n\pi))$$

$$r^5 (\cos 5\theta + i \sin 5\theta) = 1(\cos 2n\pi + i \sin 2n\pi)$$

$$r^5 = 1$$

$$5\theta = 2n\pi$$

$$r = 1$$

$$\theta = \frac{2n\pi}{5}$$

$$\text{let } n = 0, 1, 2, 3, 4$$

$$\theta = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{8\pi}{5}$$

$$= 0, \frac{2\pi}{5}, \frac{4\pi}{5}, -\frac{4\pi}{5}, -\frac{2\pi}{5}$$

$$z_1 = \cos 0 + i \sin 0$$

$$= 1$$

$$z_2 = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$$

$$= w$$

$$z_3 = \cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5}$$

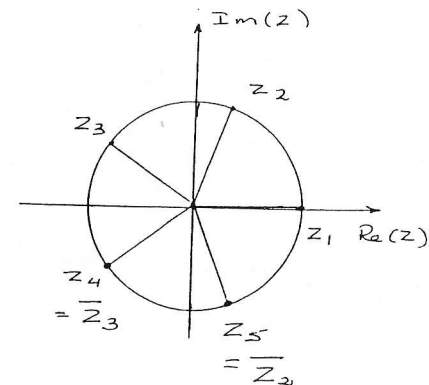
$$= w^2$$

$$z_4 = \cos \frac{6\pi}{5} + i \sin \frac{6\pi}{5}$$

$$= w^3$$

$$z_5 = \cos \frac{8\pi}{5} + i \sin \frac{8\pi}{5}$$

$$= w^4$$



$$z_4 = \cos \frac{6\pi}{5} + i \sin \frac{6\pi}{5}$$

$$= \cos\left(-\frac{4\pi}{5}\right) + i \sin\left(-\frac{4\pi}{5}\right)$$

$$= \cos \frac{4\pi}{5} - i \sin \frac{4\pi}{5}$$

$$= \bar{z}_3$$

$$z_5 = \cos \frac{8\pi}{5} + i \sin \frac{8\pi}{5}$$

$$= \cos\left(-\frac{2\pi}{5}\right) + i \sin\left(-\frac{2\pi}{5}\right)$$

$$= \cos \frac{2\pi}{5} - i \sin \frac{2\pi}{5}$$

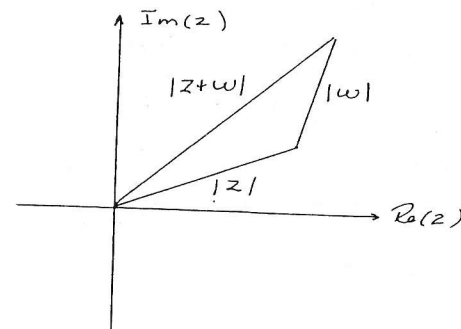
$$= \bar{z}_2$$

$$(iv) z_1 + z_2 + z_3 + z_4 + z_5 = -\frac{b}{a}$$

$$1 + w + w^2 + w^3 + w^4 = 0$$

$$\therefore w + w^2 + w^3 + w^4 = -1$$

b)



TRIANGLE INEQUALITY

$$|z + w| \leq |z| + |w|$$

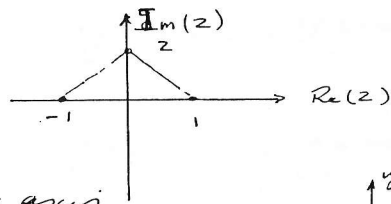
c)

$$(i) \left| \frac{z+1}{z-1} \right| = 1$$

$$\frac{|z+1|}{|z-1|} = 1$$

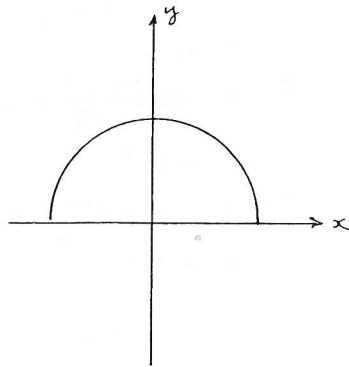
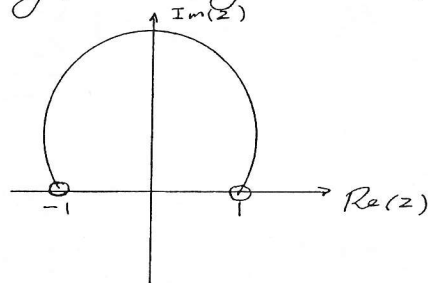
$$|z+1| = |z-1|$$

Locus is the y axis.



$$(ii) \arg \left| \frac{z+1}{z-1} \right| = \frac{\pi}{4}$$

$$\arg(z+1) - \arg(z-1) = \frac{\pi}{4}$$



$$d) \frac{(\cos 50^\circ - i \sin 50^\circ)^2 (\cos 30^\circ + i \sin 30^\circ)^8}{(\cos 20^\circ + i \sin 20^\circ)^5}$$

$$= \frac{(\cos(-50^\circ) + i \sin(-50^\circ))^2 (\cos 30^\circ + i \sin 30^\circ)^8}{(\cos 20^\circ + i \sin 20^\circ)^5}$$

$$= \frac{(\cos 0^\circ + i \sin 0^\circ)^{-5} (\cos 0^\circ + i \sin 0^\circ)^{24}}{(\cos 0^\circ + i \sin 0^\circ)^{10}}$$

$$= \frac{(\cos 0^\circ + i \sin 0^\circ)^{-10} (\cos 0^\circ + i \sin 0^\circ)^{24}}{(\cos 0^\circ + i \sin 0^\circ)^{10}}$$

$$= (\cos 0^\circ + i \sin 0^\circ)^4$$

$$= \cos 40^\circ + i \sin 40^\circ$$